Answer Key - Mathematics - K-12 Level Assessment 8 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) 7; B) 3; C) 12; D) 4
   Note: Correct option: B - 3
2. **Answer:** C
   Options: A) 0; B) 3; C) 9; D) 12
   Note: Correct option: C - 9
3. **Answer:** C
   Options: A) 2 km; B) 2 cm; C) 2 m; D) 2 mm
   Note: Correct option: C - 2 m
4. **Answer:** D
   Options: A) 0.5; B) 0; C) -1; D) -1.5
   Note: Correct option: D - -1.5
5. **Answer:** B
   Options: A) 1; B) -3; C) -1; D) 3
   Note: Correct option: B - -3

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.
7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '0.261'.
9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** By Vieta's formulas, $ab + ac + bc = 5$ and $abc = -7$, so

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{ab + ac + bc}{abc} = -\frac{5}{7}.$$

    Note: Students should show all steps in their mathematical solution.

12. **Answer:** m = e/6. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key